



GROWTH CLOUD

Project Documentation

An AI-powered marketing, lead management, automation and analytics platform



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01. PROJECT AT A GLANCE

Growth Cloud is an API-first, multi-tenant marketing automation platform designed for businesses to capture leads, automate marketing workflows, send email campaigns, score leads and analyze campaign performance.

It brings lead management, email marketing, workflow automation, segmentation, analytics, AI assistance, webhooks and API integrations into one centralized workspace.

Core Modules

- Lead Management
 - Email Marketing & Campaigns
 - Workflow Automation
 - Lead Scoring
 - Segmentation
 - Analytics
 - AI Marketing Assistant
 - Webhooks & API Integration
 - Workspace Management
 - Integrations
-

02. PROBLEM, USERS & PRODUCT VALUE

Problem

Businesses often use multiple tools for lead capture, email campaigns, customer engagement, automation and analytics.

This creates fragmented data and makes it difficult to track leads, automate follow-ups and understand campaign performance.

Growth Cloud brings these marketing workflows together in one platform.

Target Users

Marketers

Create campaigns, manage audiences, automate communication and track performance.

Developers

Use APIs, SDKs and webhooks to integrate Growth Cloud into existing applications.

Business Owners

Monitor leads, conversions, campaign performance and overall marketing activity.

Agencies

Manage marketing operations and customer engagement across multiple workspaces.

Product Value

Unified marketing operations + Lead management + Automation + Analytics + AI assistance

03. ARCHITECTURE & TECHNOLOGY

Growth Cloud uses an API-first, multi-tenant architecture with a separate frontend, backend and data layer.

Frontend: Next.js + React + Tailwind CSS

Backend: Node.js + Express + TypeScript

Database: PostgreSQL

Queue: Redis + BullMQ

Authentication: JWT Authentication

AI: OpenAI API

Email: SendGrid / Amazon SES

Integrations: APIs, SDKs & Webhooks

The project work log identifies Next.js and React on the frontend, Node.js/Express on the backend, PostgreSQL for core data, Redis and BullMQ for queues, JWT authentication, OpenAI for AI functionality, and email providers such as SendGrid or Amazon SES.

Architecture

User → Next.js → REST API → Express → PostgreSQL

Background Jobs → Redis → BullMQ → Email / Workflow Processing

04. CORE CRM WORKFLOW

The main Growth Cloud workflow connects lead capture, scoring, segmentation, campaigns, automation and analytics.

Landing Page → Authentication → Workspace → Dashboard → Leads → Scoring → Segments → Campaigns → Workflows → Analytics / AI

Lead Workflow

Lead → Score → Activity → Segment → Campaign → Engagement → Conversion

Campaign Workflow

Segment → Email → Queue → Email Provider → Tracking → Analytics

Automation Workflow

Trigger → Condition → Action → Queue → Workflow Execution → Analytics

The final project flow includes leads, campaigns, workflows, segments, scoring, analytics, AI Assistant, integrations and settings as the main workspace areas.

05. FEATURE MAP

Dashboard

Provides an operational overview of leads, campaigns, workflows, analytics and marketing activity.

Leads

Manage leads, tags, custom fields, assignments, activities, notes, lifecycle stages and consent.

Campaigns

Create and manage email campaigns, templates, personalization, scheduling and delivery.

Workflows

Create automated marketing journeys using triggers, conditions, delays and actions.

Segments

Create dynamic and static audience segments using filters, behavior and lead scores.

Lead Scoring

Define positive and negative scoring rules, monitor score history and automatically recalculate lead scores.

Analytics

Track lead, campaign, email, workflow and source performance through charts, metrics and filters.

AI Assistant

Generate emails, summarize leads, assist with scoring, create segments and provide campaign insights.

Integrations

Manage API keys, SDK functionality, webhooks and external integrations.

06. C0 WEB CHAT

C0 Web Chat was used during the initial research and planning stage of Growth Cloud.

It was mainly used to understand the problem statement and convert the broad product requirements into structured technical documentation.

C0 Usage

Product research

Problem analysis

Requirement planning

PRD generation

TRD generation

Architecture planning

Feature breakdown

Implementation planning

Workflow

Problem Statement → C0 Research → PRD / TRD → Build Design → Build Prototype → CORA Implementation

The work log specifically records that C0 was used to research the problem statement and generate the project plan along with the PRD and TRD.

07. C0 PROMPT HIGHLIGHTS

C0 was mainly used for:

- Problem research
- Product planning
- PRD generation
- TRD generation
- Architecture analysis
- Feature planning
- Workflow planning
- Implementation direction

Planning Areas

Foundation → Authentication → Workspace → Leads → Campaigns → Workflows → Segments → Scoring → Analytics → AI → Integrations

Prompts Used

Prompt 1 : Problem & Product Research

Analyze the Growth Cloud problem statement and research the key challenges businesses face when managing leads, email campaigns, marketing automation, segmentation and analytics across multiple disconnected tools. Identify the target users, their major pain points, the core product value and the workflows that should be unified inside an API-first marketing automation platform.

Prompt 2 : PRD Planning

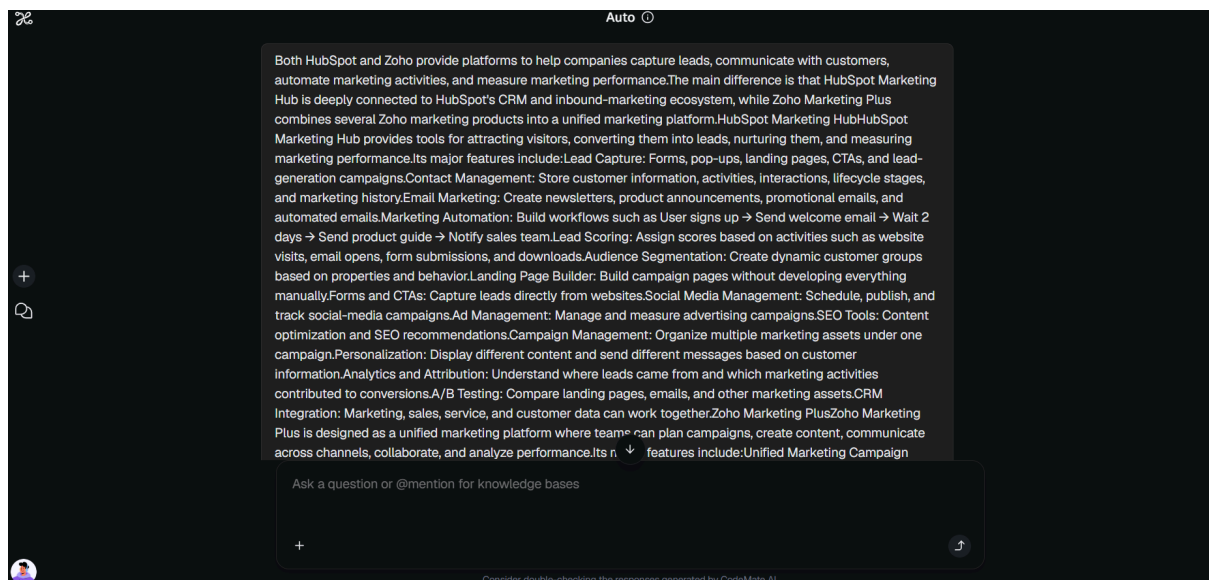
Create a detailed Product Requirements Document for Growth Cloud, an API-first multi-tenant marketing automation platform. Cover the product overview, target users, lead management, email campaigns, workflows, segmentation, lead scoring, analytics, AI features, integrations, webhooks, workspace management and user roles. Structure the requirements so they can later be converted into clear development tasks.

Prompt 3 : TRD & Architecture Planning

Create a Technical Requirements Document for Growth Cloud based on the defined product requirements. Break down the frontend, backend, database, authentication, workspace isolation, API structure, background jobs, Redis queues, email delivery, webhooks, analytics, AI services and integrations. Define how these components should communicate and how the system should support a scalable multi-tenant architecture.

Prompt 4 : Implementation Planning

Review the Growth Cloud PRD and TRD and convert the product requirements into an incremental development plan. Prioritize authentication and workspace setup first, followed by lead management, email marketing, workflows, segmentation, scoring, analytics, webhooks, AI features and SDK integrations. Identify dependencies between modules and organize the implementation so each major stage can be tested before moving to the next.



08. BUILD: DESIGN MODE + PROTOTYPE MODE

Design Mode

Build Design Mode was used to establish the initial visual direction and product experience for Growth Cloud.

The design focused on:

- Modern B2B SaaS interface
- Clean marketing experience
- Dashboard-focused product UI
- Lead management
- Campaign workflows
- Automation
- Analytics
- AI-powered functionality
- Responsive layouts
- Consistent reusable components

The Design Mode phase was started after the initial C0 research and planning stage.

Prompts Used

Prompt 1 : Initial Product Design

Design a modern production-ready B2B SaaS platform called Growth Cloud, an AI-powered marketing automation and lead management platform for businesses. Create a professional product experience covering lead management, email marketing, workflow automation, segmentation, analytics, AI assistance and integrations. The interface should feel like a real production SaaS product with clear navigation, strong information hierarchy, reusable components and responsive layouts.

Prompt 2 : Dashboard & Marketing Workspace

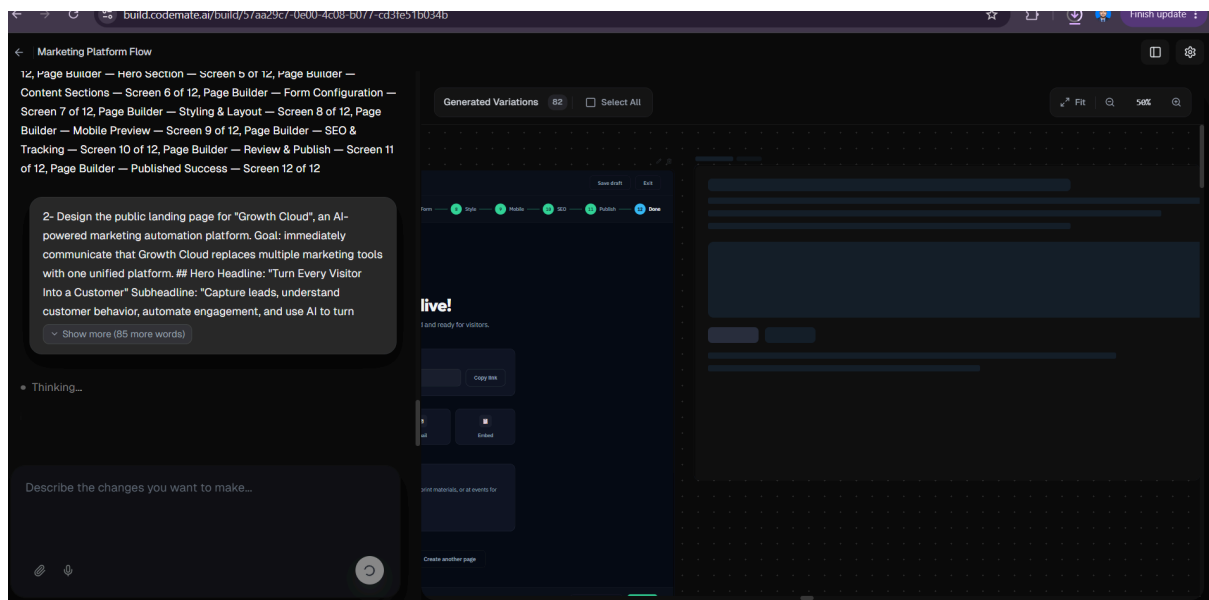
Design the Growth Cloud dashboard as the central marketing operations workspace. Include lead metrics, campaign performance, conversion trends, workflow activity, source analytics, score distribution and AI-generated insights. Create clear navigation for Leads, Campaigns, Workflows, Segments, Scoring, Analytics, AI Assistant, Integrations and Settings. Keep the interface clean, data-focused and easy to scan.

Prompt 3 : CRM & Campaign Experience

Create detailed product interfaces for lead management and email marketing. Include searchable and filterable lead tables, lead profiles, activity timelines, tags, custom fields, lifecycle stages, campaigns, templates, personalization, scheduling, email tracking and campaign statistics. Use reusable components and maintain a consistent Growth Cloud design system across all screens.

Prompt 4 : Automation & Analytics

Design the Growth Cloud workflow builder and analytics experience. The workflow builder should support triggers, conditions, delays and actions with clear execution states and logs. The analytics section should provide lead, campaign, email, workflow and source metrics with charts, date filters, exports and AI-generated insights. Make the experience responsive and consistent with the rest of the product.



Pototype Mode

The generated Growth Cloud design was then taken forward into application prototyping and development.

The prototype stage was started after the Design Mode work and was used to create the application structure and core product flows.

Prompts Used

Prompt 1 : Convert Design into Application

The Growth Cloud design has been finalized. Do not redesign or change the approved layout, typography, colors, spacing, navigation or component structure. Now convert the existing design into a functional marketing automation application with reusable components, responsive layouts and realistic product interactions. Implement the main workspace, dashboard, leads, campaigns, workflows, segments, scoring, analytics, AI Assistant, integrations and settings.

Prompt 2 : Backend Integration

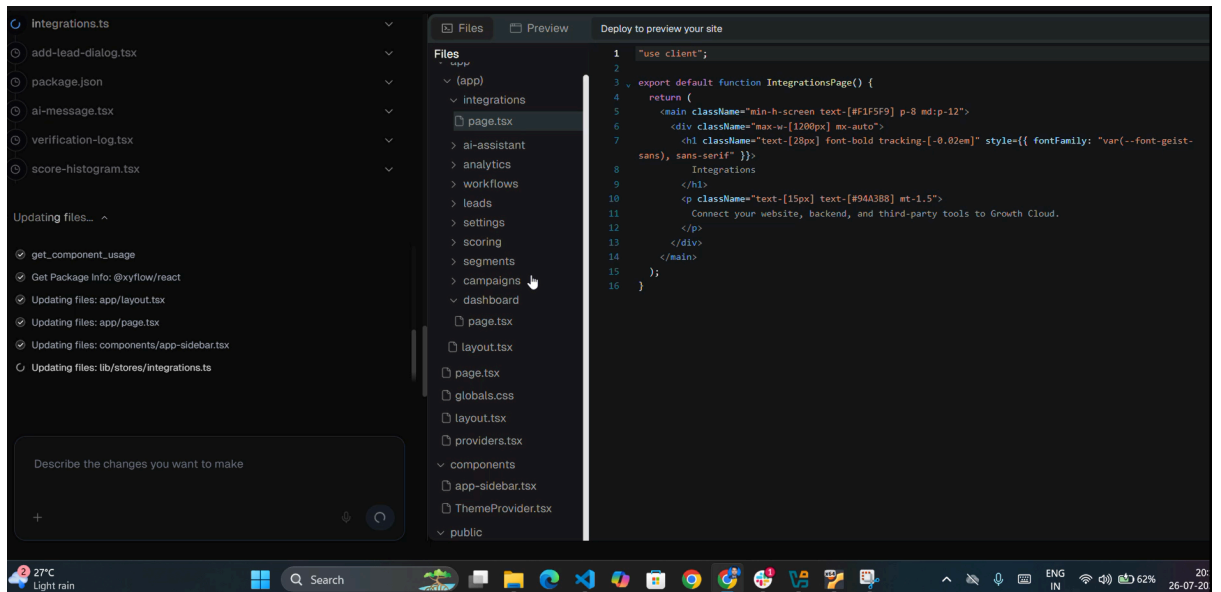
Continue the existing Growth Cloud implementation without changing the approved UI. Connect the frontend with the backend API and PostgreSQL database. Replace static or mock data with persistent application data and establish the required API communication for authentication, workspaces, leads, campaigns, workflows, segments, scoring and analytics. Preserve the current frontend architecture and ensure the API responses are handled correctly.

Prompt 3 : Core Marketing Functionality

Implement the core Growth Cloud marketing functionality including lead CRUD, tags, custom fields, assignments, activity timelines, notes, CSV import/export, search, filtering, sorting, pagination and lifecycle stages. Then connect the implementation to the frontend and ensure the lead workflow integrates correctly with campaigns, segments, scoring and automation.

Prompt 4 : Email & Campaign Functionality

Continue development of the Growth Cloud email marketing module. Implement email templates, variables, personalization, campaign scheduling, single send, bulk send, email queues, open and click tracking, bounce handling, spam handling, unsubscribe and suppression lists. Integrate the supported email providers while preserving the existing product UI and ensuring the complete campaign workflow works end-to-end.



09. BUILD PROMPT HIGHLIGHTS

The Build prompts focused on creating a complete marketing automation product rather than a generic SaaS dashboard.

Main Prompt Focus

AI-powered marketing

Lead management

Email campaigns

Workflow automation

Segmentation

Lead scoring

Analytics

API-first architecture

Responsive product experience

Prompts Used

Prompt 1 : Product Foundation

Build a production-ready B2B SaaS marketing automation platform called Growth Cloud. The platform should combine lead capture, lead management, email marketing, workflow automation, lead scoring, segmentation, analytics, AI assistance, webhooks and API integrations into one connected workspace.

Prompt 2 : Lead Management

Build the Growth Cloud lead management experience with complete CRUD operations, tags, custom fields, assignment, activity timeline, notes, CSV import/export, search, filtering, sorting, pagination, lifecycle stages and consent tracking. Ensure leads are connected with scoring, segments, campaigns and workflow automation.

Prompt 3 : Workflow Automation

Implement a workflow automation system where users can create triggers, conditions, delays and actions. Support workflow execution, retry mechanisms, logs, history and background processing through Redis and BullMQ. Ensure workflow events can interact with leads, campaigns and scoring.

Prompt 4 : Analytics & AI

Build the analytics and AI layer for Growth Cloud. Provide lead, campaign, email, source, workflow and score metrics with charts, date filters and exports. Add AI-powered email generation, lead summaries, lead scoring, campaign insights and segment building while keeping AI connected to the relevant workspace data.

10. CORA: PLANNING & IMPLEMENTATION

CORA was used as the main implementation and development layer after the initial research and Build stages.

It was used to implement the backend structure, authentication, database, APIs, CRM/marketing modules, workflow automation, email infrastructure, analytics, integrations and AI functionality.

CORA was used for:

- Backend structure
- PostgreSQL
- Authentication
- RBAC
- Workspace management
- Lead management
- Email marketing
- Campaigns
- Workflows
- Segmentation
- Lead scoring
- Analytics
- Webhooks
- AI features
- SDK
- Bug fixes
- Testing

Workflow

Requirement → CORA Planning → Implementation → API Integration → Testing → Fixes

The work log shows the development progressing from backend structure and authentication into API integration, leads, email marketing, workflows, segmentation, scoring, analytics, webhooks, AI and SDK implementation.

Prompts Used

Prompt 1 : Backend & Authentication

Analyze the existing Growth Cloud application and implement the backend foundation required for a multi-tenant marketing automation platform. Set up the backend structure, authentication, JWT flow, workspace creation, team invitations, RBAC, session management and workspace isolation. Preserve the existing frontend and integrate the backend incrementally without breaking the current application.

Prompt 2 : Lead Management

Implement the complete Growth Cloud lead management module including CRUD operations, tags, custom fields, assignment, activity timeline, notes, CSV import/export, search, filtering, sorting, pagination, lifecycle stages and consent tracking. Connect the backend APIs to the existing frontend and ensure all lead data is persisted correctly in PostgreSQL.

Prompt 3 : Workflow & Automation

Implement the Growth Cloud workflow engine with workflow CRUD, triggers, conditions, delays, actions, workflow execution, retry mechanisms, logs and execution history. Use Redis and BullMQ for background processing and ensure workflow events can trigger the appropriate lead, campaign and scoring actions. Verify the workflow execution and error-handling behavior.

Prompt 4 : AI, Analytics & Integrations

Implement the remaining Growth Cloud platform modules including analytics, webhooks, AI features and SDK functionality. Add lead, campaign, email, source, workflow and score analytics, webhook subscriptions and retries, AI email generation, lead summaries, AI scoring, campaign insights and segment building. Implement SDK functionality for user identification, event tracking, lead synchronization and forms while keeping all features scoped to the correct workspace.

page.tsx ...api-keys 7TS email.service.test.tsapi-key-management-plan.mdintegrations-tabs.tsx 9+ X

components > integrations-tabs.tsx > integrationsTabs

```
66 export function IntegrationsTabs() {
90   const loadData = React.useCallback(async () => {
96     const nextWorkspaces = workspaceResponse.data.items;
97     const nextApiKeys = apiKeyResponse.data.items;
98
99     setWorkspaces(nextWorkspaces);
100    setApiKeys(nextApiKeys);
101
102    const nextWorkspaceId = selectedWorkspaceId || nextWorkspaces[0]?.id || "";
103    setSelectedWorkspaceId(nextWorkspaceId);
104
105    if (nextWorkspaceId) {
106      const membersResponse = await getWorkspaceMembers(nextWorkspaceId);
107      setMembers(membersResponse.data.items);
108    } else {
109      setMembers([]);
110    }
111
112    const { publicKey } = groupApiKeys(nextApiKeys);
113    setAllowedDomains(publicKey ? ["localhost", "app.localhost"] : []);
114  } catch (loadError) {
115    setError(loadError instanceof Error ? loadError.message : "Failed to load integration
116  } finally {
117    setLoading(false);
118  }
119 }, [selectedWorkspaceId]);
120
121 React.useEffect(() => {
122   void loadData();
123 }, [loadData]);
124
125 React.useEffect(() => {
126   if (!selectedWorkspaceId) return;
127
128   let active = true;
129
```

CHAT CORA

Implement Mo... Implement Mo... check what already... Implement M +

Current state:

- The app sidebar sizing/layout is aligned again.
- The frontend still builds cleanly after the layout adjustment.

check the interGr
Growth Cloud
Acme Marketing
acme.growthcloud.app
Platform
Dashboard
Leads
Campaigns...

Working (34s)

Context Condensed 251,128 — 28,278 tokens

Analyzing integrations layout and sidebar spacing.

Checkpoint

Read 3 files page.tsx, app-sidebar.tsx ...

Inspecting the integrations page and app sidebar layout structure.

Checkpoint (Current)

Applying components/app-sidebar.tsx

Todos 0/3

Write or edit code...

Code Standard

11. CORA TASK ARCHIVE

Instead of listing every individual CORA task, the implementation can be grouped into the major areas below.

Backend Foundation

Backend structure, Express REST API, PostgreSQL, database setup and service architecture.

Authentication & RBAC

Signup, login, logout, email verification, password reset, refresh tokens, JWT authentication, workspace creation, invitations, sessions and role-based access.

Lead Management

CRUD, tags, custom fields, assignment, activities, notes, CSV import/export, search, filtering, sorting, pagination, lifecycle stages and consent tracking.

Email Marketing

Templates, variables, personalization, scheduling, single send, bulk send, email queues, tracking, bounce handling, spam handling, unsubscribe and suppression lists.

Workflow Automation

Triggers, conditions, delays, actions, execution, retries, logs, history, Redis queues and BullMQ workers.

Segmentation

Dynamic segments, static segments, saved filters, score-based segments, behavior-based segments, audience previews and reusable segments.

Lead Scoring

Positive rules, negative rules, score history, threshold alerts, automatic recalculation, event-driven scoring and AI-assisted scoring.

Analytics

Lead metrics, campaign metrics, email metrics, source analytics, workflow analytics, score distribution, charts, date filters, exports and AI insights.

Webhooks

CRUD endpoints, event subscriptions, signed webhooks, retry mechanisms, delivery logs and failed-event replay.

AI

AI email generation, lead summaries, AI lead scoring, campaign insights and AI segment building.

SDK

API client, user identification, event tracking, lead sync, forms, public key authentication, documentation and examples.

The actual work log records these implementation areas across the development timeline.

12. IMPLEMENTATION MILESTONES

Phase 1: Research & Planning

Used C0 to research the problem statement and generate the PRD and TRD.

Phase 2: Design

Started the Growth Cloud interface using Build Design Mode.

Phase 3: Prototyping

Created the initial application prototype using Build.

Phase 4: Backend Foundation

Created the backend structure and initialized the PostgreSQL development environment.

Phase 5: Authentication

Implemented signup, login, logout, email verification, password reset, refresh tokens, JWT authentication, workspace creation, invitations and RBAC.

Phase 6: API Integration

Started connecting the frontend and backend APIs.

Phase 7: Lead Management

Implemented lead CRUD, tags, custom fields, assignment, activities, notes, CSV workflows, search, filters, sorting, pagination and lifecycle stages.

Phase 8: Email Marketing

Implemented templates, campaigns, personalization, scheduling, sending, queues, tracking, unsubscribe and email-provider integrations.

Phase 9: Automation & Segmentation

Implemented workflows, triggers, conditions, delays, actions, Redis/BullMQ execution, dynamic segments and behavior-based segmentation.

Phase 10: Scoring & Analytics

Implemented lead scoring, score history, threshold alerts, campaign analytics, workflow analytics, source analytics and AI insights.

Phase 11: Integrations & AI

Implemented webhooks, SDK functionality and AI-powered marketing features.

Phase 12: Testing & Refinement

Implemented test scripts and resolved issues across lead creation, workflow triggers, analytics data fetching, email functionality and service dispatch.

13. BACKEND, DATA & INTEGRATIONS

Growth Cloud uses a Node.js / Express REST API connected to PostgreSQL.

Core Data

- Users
- Workspaces
- Workspace Members
- Leads
- Lead Activities
- Tags
- Custom Fields
- Campaigns
- Email Templates
- Campaign Recipients
- Segments
- Score Rules
- Workflows
- Workflow Runs
- Analytics
- Webhooks
- API Keys
- Notifications
- AI Data

Backend Modules

- Auth
- Workspaces
- API Keys
- Leads
- Email
- Segments
- Scoring
- Workflows
- Webhooks
- SDK
- Analytics
- AI

The final project summary confirms these backend modules and the `/api/v1` API structure.

Integrations

Redis / BullMQ → Background jobs

SendGrid / Amazon SES → Email delivery

OpenAI → AI features

Webhooks → Third-party integrations

SDK → Product tracking and lead synchronization

Application Flow

Frontend → REST API → Backend Services → PostgreSQL

Workflow / Email Events → Redis → BullMQ → Background Processing

14. AI ASSISTANT & WORKSPACE CONTEXT

The AI Assistant is integrated into Growth Cloud as part of the marketing workflow rather than functioning as a standalone chatbot.

AI Capabilities

- AI Email Generator
- AI Lead Summary
- AI Lead Scoring
- AI Campaign Insights
- AI Segment Builder

These AI features were implemented as part of the Growth Cloud development process.

AI Flow

User Request → Workspace Context → Leads / Campaigns / Workflows → AI Processing → Response / Recommendation

AI Assistant

The AI Assistant can help users understand and work with marketing data, generate campaign content, summarize leads, assist with scoring and provide campaign insights.

Workspace Context

AI functionality should remain connected to the appropriate workspace so that marketing information is used within the correct product context.

The final application flow specifically connects the AI Assistant with leads, campaigns and workflows.

15. RESPONSIVE UI & POLISH

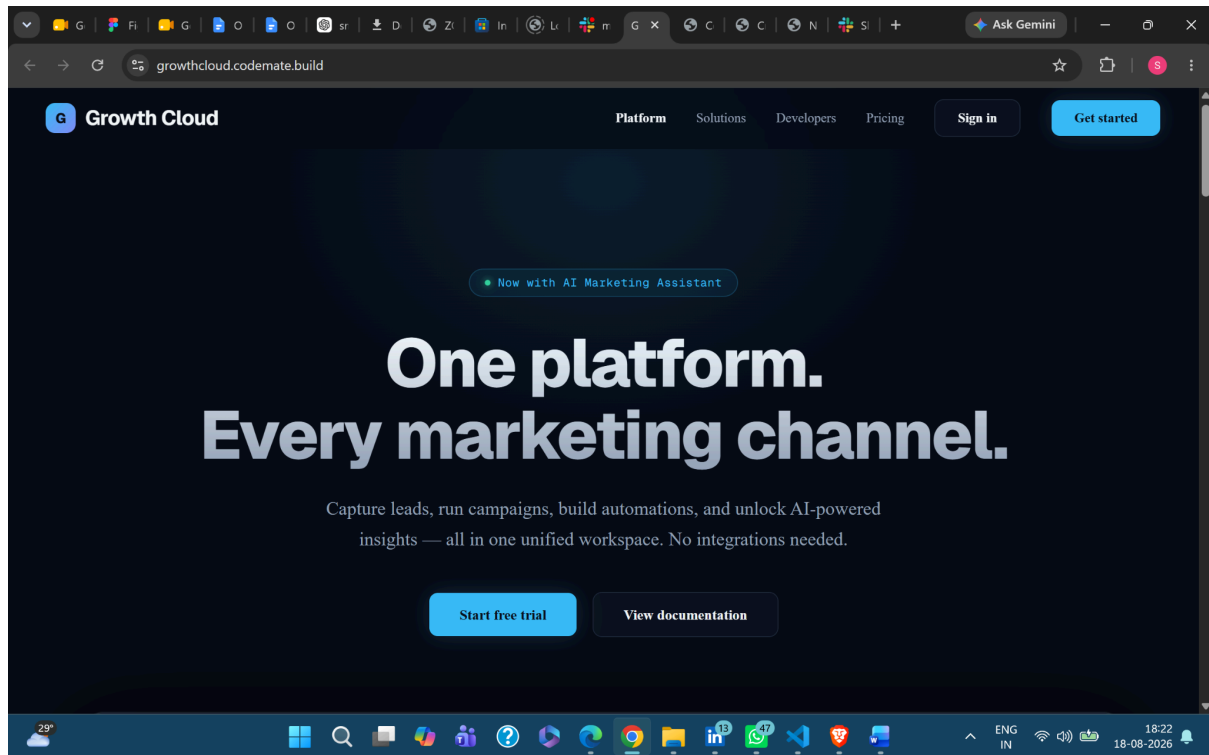
Growth Cloud was designed as a modern B2B SaaS platform with responsive layouts across the marketing website and application workspace.

UI Characteristics

- Clean SaaS interface
- Consistent spacing
- Reusable components
- Professional typography
- Responsive layouts
- Data-focused dashboards
- Tables and filters
- Charts
- Forms
- Modals
- Status indicators
- Loading states
- Error states
- Empty states
- Mobile-friendly layouts

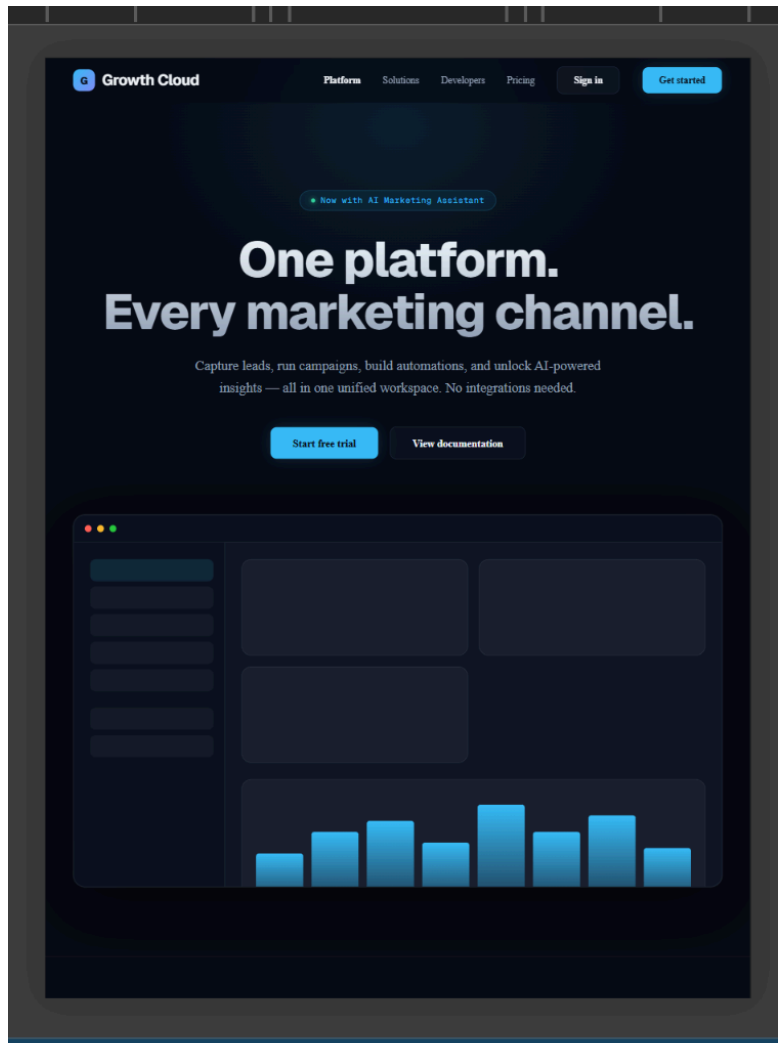
Desktop

Full dashboard experience with navigation, tables, analytics, workflow builders and multi-column layouts.



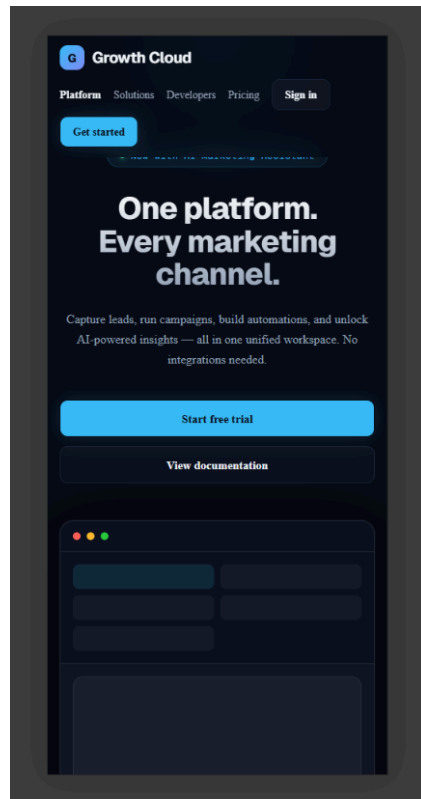
Tablet

Adjusted content density and navigation while preserving the main workspace experience.



Mobile

Stacked layouts, simplified navigation and touch-friendly interactions.



Final Product Flow

C0 Web Chat → Build Design Mode → Build Prototyping → CORA Development → Final Product Preview

The work log confirms this overall progression from C0 research and planning, to Design Mode, Build prototyping, backend development and final implementation.
